Support-aware digit coding and sharper Pell norms

Explicit Diophantine equations for the consistency of ZFC

Research synthesis prepared for Vladimir Reshetnikov
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Abstract

We give explicit, fixed-integer-coefficient polynomial equations whose absence of solutions is provably equivalent in ZFC to its syntactic consistency, using the repository's established K215/compiler calibration. The principal new step replaces a full Sidon condition in quadratic digit coding by isolation only of monomials actually used by the source equations. A certified list of 32 labels in $[0, 200]$ isolates all 71 active pairs, although many of the other 528 pairs collide. Compared with the wave-6 constructions at the pinned source commit, this gives nine positive unknowns at degree 32 568 568 in 652 binary arithmetic operations, and ten unrestricted integer unknowns at degree 59 564 744 in 653 operations. The latter uses Sun's cited ten-integer bridge. A separate root estimate permits index-quotient absorption and the removal of an inner radical separator to coexist: eleven positive unknowns then suffice at degree 990 in 117 operations, or degree 994 in 115 operations. A small, historically qualified circuit refinement also attains degree 44 with fifteen positive unknowns in 76 operations. All coefficients, arithmetic circuits, finite certificates, reconstruction scripts and source notices accompany the article. These are ordinary mathematical proofs with explicit inherited interfaces, not a new end-to-end proof-assistant certificate, a proof of consistency, or a claim of absolute minimality.

Witness domain	Exact degree	Operations	Role
$\mathbb{Z}_{>0}^{9}$	32 568 568	652	Few positive variables
$\mathbb{N}^{9}$	32 568 568	655	Natural-domain translation
$\mathbb{Z}^{10}$	59 564 744	653	Unrestricted integer equation
$\mathbb{Z}_{>0}^{11}$	990	117	Moderate-degree construction
$\mathbb{Z}_{>0}^{11}$	994	115	Shorter moderate-degree evaluator
$\mathbb{Z}_{>0}^{15}$	44	76	Small low-degree refinement

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1 Scope, baseline and the exact claim

1.1 What "simplest" can mean

Variable count, degree, evaluator length and coefficient size are different objectives. No known argument identifies a globally simplest consistency equation under all of them. This package constructs improvements with precise, comparable metrics. The underlying logical assertion is

$$\mathsf{ZFC} \vdash (\neg\exists \mathbf{x} \in \mathcal{A}\ P(\mathbf{x}) = 0) \iff \mathrm{Con}(\mathsf{ZFC}), \tag{1}$$

where $\mathcal{A}$ is stated separately for every polynomial. Here $\mathbb{N} = \{0, 1, 2, \ldots\}$, and $\mathrm{Con}(\mathsf{ZFC})$ means that the repository's fixed, recursively specified ZFC proof calculus has no finite derivation of falsity. No assertion is made that a positive-domain polynomial has the same meaning over all integers.

The source is pinned to `124c07c0b77a9e20e84676514d8b1ff78536ea3f`. This commit places wave 6 beside wave 5; the collection README at that commit still principally describes wave 5. Consequently the comparison uses the actual new packages 38–45 as well as the active research documents. A comparison only with wave 5 would substantially overstate the improvements [1, 4, 5, 6, 7].

Comparable construction	Variables	Degree	Operations
Package 45 multi-target	$9\mathbb{Z}_{>0}$	129 875 980	788
This package	$9\mathbb{Z}_{>0}$	32 568 568	652
Package 40 quadratic packing	$10\mathbb{Z}$	237 520 456	782
This package	$10\mathbb{Z}$	59 564 744	653
Package 38 absorption	$11\mathbb{Z}_{>0}$	1154	123
This package, full carrier	$11\mathbb{Z}_{>0}$	990	117
This package, short carrier	$11\mathbb{Z}_{>0}$	994	115
Historical wave-2 gap SOS	$15\mathbb{Z}_{>0}$	44	78
This package, regrouped SOS	$15\mathbb{Z}_{>0}$	44	76

Table 1: Counts are for supplied circuits, not circuit lower bounds. The degree-44 idea is historical; only its shorter regrouping is new here.

The nine-variable degree decreases by approximately 74.92% and its operation count by 17.26%. The ten-integer degree decreases by approximately 74.92% and its operation count by 16.50% against package 40. These are substantial simplifications of explicit instances, not new general nine- or ten-variable representation theorems. The historical 67-operation equation remains shorter when evaluator length alone is the objective. The new constructions do not decrease the known variable-count bounds.

1.2 Proof layers and what was checked

The mathematical composition has four layers:

1. the literal K215 machine halts on blank input exactly when there is a finite ZFC refutation;
2. its marked-tile/native-window/Pell compiler gives the fixed positive source equations and coherent coefficient tuple;
3. the new support-aware coding or the new eleven-variable root estimate transforms those equations;

4. the existing Jones ratio or Sun Lucas bridge, where used, and an explicit finalizer give a single polynomial.

Layers 1 and 2 are inherited mathematical dependencies. Their source proofs and immutable repository references are identified in the archive. The new local arguments are proved here. The finite programs check literal coefficient reconstruction, symbolic extraction identities, arithmetic circuits, liveness and degree certificates. They cannot replace the unbounded compiler arguments or decide either side of (1).

The repository separately contains an unconditional Lean equivalence with a nine-natural-variable polynomial selected through `Classical.choose`. That milestone has different coefficient explicitness from the materialized constructions here. A Lean theorem about a syntactic proof predicate also differs from an already checked derivation inside the object-language ZFC calculus. We do not conflate these claims or claim a new Lean/Rocq build [2].

1.3 Exact coefficients and operation convention

All constructions use the *whole* package-31 coefficient tuple (β, K_0, M_C, M_F), where M_F is already shifted. Its bit lengths are

$$1064921, \quad 1070004, \quad 1064921, \quad 1064922.$$

The four raw payloads total 533099 bytes; their gzip encodings total 10485 bytes. They are independently reconstructed from the finite recipe in Appendix A. The recipe does not search for a refutation or depend on a truth value of consistency.

An operation is one binary addition, subtraction or multiplication of a variable-dependent expression, including multiplication by a numeral. Reuse and fixed coefficient leaves are free. Fixed powers are expanded into charged multiplication gates. Arithmetic involving only fixed constants is part of coefficient construction, as in the compared packages. Equality comparisons are free when reporting systems. The literal DAGs, rather than uncharged abbreviations, determine all counts.

2 The quadratic source and support-aware coding

2.1 The complete parent system

The common source has 31 positive coordinates, in this exact order:

$$J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,$$
$$N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta.$$

These names are local to this subsection. Put $q = \beta J + 1$, $k = \eta + \zeta$, $H_p = 4a + 3$ and $C_p = q - F - \alpha$. The inherited source is the following quadratic system [3, 5].

Row	Equation
1	$L = q^2 - 1$
2	$N = q(L + 1)$
3	$X = wq$
4	$a = E + sN$
5	$E = X(a - E)$

Row	Equation
6	$c = k(a - E) + \eta$
7	$P = E(c - \eta) + k$
8	$\tau^2 = P(P - k) + 1$
9	$D = X + ac + \gamma H_p$
10	$\Delta = a^2 + H_p$
11	$C_2 = c^2$
12	$D^2 = \Delta C_2 + 1$
13	$k = jc - V + 1 + hE$
14	$jc - V = (q - F)(N - q) - C_p L + (M_C + qM_F)J$
15	$(K_0 + w)C_p + q - F - z(q - 1) = 1$
16	$F_2 = f^2$
17	$V = of - c$
18	$g = y^2 - V^2$
19	$I = iC_2$
20	$B_{\mathrm{aux}} = I^2$
21	$B_{\mathrm{aux}} = \Delta(F_2 - 1)$
22	$y^2 - B_{\mathrm{aux}} g = 1$

The ten-integer construction uses this table. The nine-variable construction uses the inherited positive transport shift in row 15,

$$(K_0 + w)C_p = F + z(q - 1).$$

It corresponds to $z_{\mathrm{old}} = z + 1$. Canonical compiler witnesses have $z_{\mathrm{old}} \geq 2$, so this is an existence equivalence on the positive domain. Both parent conventions and their shift are checked symbolically; their monomial supports coincide. They are not treated as literally the same polynomial system at the same coordinates.

For each parent, replace all coordinates by $x_i + 1$, form left-minus-right residuals, and reorder the rows as $1, 2, 3, 4, 6, 7, \ldots, 22, 5$. Write the result as $p_1, \ldots, p_{22} \in \mathbb{Z}[x_1, \ldots, x_{31}]$. Each has degree at most two, and the last has $p_{22}(0) = 1$. Thus the all-zero tuple is not a common zero. The inherited theorem identifies a common natural zero with a finite ZFC refutation.

2.2 The weaker separation property

Pad a monomial of degree less than two by copies of $x_0 = 1$. Let $\mathcal{S}$ be the union of the unordered pairs (i, j) occurring in the padded supports of the residuals. A weight vector $w_0 = 0, w_1, \ldots, w_n \geq 0$ has *active-pair isolation* when

$$(i, j) \in \mathcal{S}, \quad w_i + w_j = w_k + w_l, \quad 0 \leq k \leq l \leq n \quad \Longrightarrow \quad (i, j) = (k, l). \tag{2}$$

The comparison is against *all* pairs, not just supported ones. Requiring uniqueness only among supported pairs would allow an unused monomial in the square of the packed word to contaminate a tested digit. Pairs outside $\mathcal{S}$ may collide with one another.

Theorem 2.1 (Support-aware simultaneous extraction)**.** *Let* $p_j = \sum_{|\alpha| \leq 2} a_{j,\alpha} x^\alpha$, $1 \leq j \leq r$, *and suppose* (2) *holds. Put* $W = \max_i w_i$ *and*

$$E_j = 2W + (j - 1)(2W + 1), \qquad c_{j,\alpha} = (2 - |\alpha|)! \prod_i \alpha_i!\, a_{j,\alpha}.$$

For

$$\mathcal{D}(Z) = \sum_{j,\alpha} c_{j,\alpha} Z^{E_j - \alpha\cdot w}, \qquad C_{\mathbf{x}}(Z) = \left(1 + \sum_i x_i Z^{w_i}\right)^2,$$

all exponents are nonnegative and

$$[Z^{E_j}] C_{\mathbf{x}}(Z)\mathcal{D}(Z) = 2p_j(\mathbf{x}).$$

If $|c_{j,\alpha}| \le M$, then for $\mathbf{x} \in \mathbb{N}^n$ every coefficient of $C_{\mathbf{x}}\mathcal{D}$ has absolute value at most $M(1+\sum_i x_i)^2$.

Proof. For a supported α, active-pair isolation gives exactly one padded pair at exponent $\alpha \cdot w$ in $C_{\mathbf{x}}$. Its coefficient is $2!x^\alpha/((2-|\alpha|)!\prod_i \alpha_i!)$. The factorial in $c_{j,\alpha}$ therefore produces $2a_{j,\alpha}x^\alpha$. A contribution from a different row would equate a nonzero target difference, of absolute value at least $2W+1$, to a difference of two pair sums, whose absolute value is at most $2W$. That is impossible. Within a row, the active monomial is isolated against every possible pair in $C_{\mathbf{x}}$, so there are no other contributions.

The reversed supports of the rows lie in disjoint intervals $[E_j - 2W, E_j]$. Each coefficient of $\mathcal{D}$ is consequently bounded by M. All coefficients of $C_{\mathbf{x}}$ are nonnegative, including those at collided unused positions, and their sum is $C_{\mathbf{x}}(1) = (1+\sum_i x_i)^2$. The coefficient bound follows by convolution. This last step is why unused collisions cause no hidden carry problem. □

2.3 The literal certificate

In the parent coordinate order above, take

$$\begin{aligned}(w_1, \ldots, w_{31}) = (&200, 199, 193, 166, 11, 57, 125, 37, 93, 113, 153, 16,\\ &158, 192, 120, 64, 139, 198, 144, 23, 15, 43, 18, 165, 179,\\ &112, 10, 44, 117, 71, 119).\end{aligned} \tag{3}$$

There are 71 active pairs among the 528 unordered pairs, including repetitions and the padding coordinate. Exhaustive enumeration verifies (2). The 528 pairs occupy only 285 sums: 130 sum positions have collisions, and the largest multiplicity is seven. Thus the condition is strictly weaker than a full Sidon condition in this instance. The certificate is the literal vector and the exhaustive enumeration; the heuristic that found it is not a premise of the proof.

Here

$$W = 200, \quad E_j = 400 + 401(j-1), \quad E_{\max} = 8821, \quad K_c = E_{\max} + 2W = 9221.$$

The previous full-Sidon vector had $W = 799$. Both parent conventions have 136 nonzero coefficient records, and the audit verifies all 44 target identities with the four native coefficient ports left as independent symbols. The coefficient bound is unchanged, because the source coefficients and the convolution estimate are unchanged.

This is a reusable sufficient-condition theorem and a specific improved certificate. We do not assert that support-aware additive coding is new in all contexts, that $W = 200$ is optimal, or that this reduction changes the undecidability variable-count records.

3 The nine-variable polynomial

3.1 Source registers and bounds

First use $\epsilon, \phi, n \in \mathbb{N}$ and $g, h, s, w, i, j \in \mathbb{Z}_{>0}$. These are exactly nine coordinates; every other name below is an abbreviation. Use the shifted parent from Section 2. The 136 factorial-weighted

coefficient records satisfy

$$|c_{r,\alpha}| < M = 2^{2134927}.$$

The verifier derives this by summing absolute integer coefficients times the powers of two supplied by the four port bit lengths. Set $b_* = 2^{2134939} = 4096M$. Define

$$\begin{aligned}
b &= b_* + \epsilon, & \mathcal{B} &= b^3, & \ell &= \sum_{r=1}^{31} \mathcal{B}^{w_r},\\
N_0 &= \mathcal{B}^{201}, & N_1 &= 4\mathcal{B}^{9222}, & N &= N_0 N_1,\\
M_c &= N_0 - 1 - (b-1)\ell, & T_* &= \sum_{r=1}^{22} \mathcal{B}^{E_r},\\
T_c &= M_c + (\mathcal{B}-2)T_* N_0,\\
J_c &= 2(1+g)^2\mathcal{D}(\mathcal{B}) + \mathcal{B}\sum_{r=0}^{9221} \mathcal{B}^r,\\
S_c &= g + J_c N_0,\\
R &= (N-1)\big((S_c + T_c + 1)N + T_c + 1\big),\\
G_c &= 32b\mathcal{B}^{200}.
\end{aligned}$$

The weights and target exponents are the fixed integers in (3); the parent residuals determine $\mathcal{D}$ exactly. All powers and sums have fixed, finite exponents and limits.

The leading coefficient of $\mathcal{D}(Z)$, at degree 8821, is 2, because the last parent residual has constant coefficient 1. All other coefficients have magnitude less than M. Since $\mathcal{B} - 1 > M$,

$$0 < \mathcal{B}^{8821} < \mathcal{D}(\mathcal{B}) < \frac{M\mathcal{B}^{8822}}{\mathcal{B}-1}. \tag{4}$$

The selected weights form a proper subset of $\{0, \dots, 200\}$, so $M_c > 0$. Consequently $R \geq N \geq 8$ and $R, N \geq b$ on every outer assignment, even before imposing the final equation.

On the branch $g < 4b\mathcal{B}^{200}$, integrality gives $1 + g \leq 4b\mathcal{B}^{200}$ and $g < N_0$. Equation (4) and $b \geq 4096M$ imply

$$2(1+g)^2\mathcal{D}(\mathcal{B}) < \frac{32Mb^2\mathcal{B}^{9222}}{\mathcal{B}-1}, \qquad 32Mb^2 \leq \mathcal{B}/128.$$

Adding the geometric sum therefore gives $0 < J_c < N_1$. Also $(\mathcal{B}-2)T_* < \mathcal{B}^{8822}$, so

$$0 \leq S_c, T_c < N. \tag{5}$$

These bounds hold on the stated range without assuming a successful digit test.

3.2 The carry equivalence

For nonnegative integers let $\tau_2(S,T) = \mathrm{pc}(S) + \mathrm{pc}(T) - \mathrm{pc}(S+T)$, where pc counts binary one-bits. This counts carries with multiplicity and is zero precisely when the binary supports are disjoint.

Lemma 3.1. *If $N = 2^e$ and $0 \leq S, T < N$, then for $R = (N-1)((S+T+1)N + T + 1)$,*

$$N^2 \mid \binom{2R}{R} \quad \Longleftrightarrow \quad \tau_2(S,T) = 0.$$

Proof. The exact identity

$$R = (S+T)N^2 + (N-1-S)N + (N-1-T)$$

has two low blocks that are binary complements. Hence $\mathrm{pc}(R) = 2e - \tau_2(S,T)$. The factorial valuation identity $\nu_2(m!) = m - \mathrm{pc}(m)$ gives $\nu_2\binom{2R}{R} = \mathrm{pc}(R)$. The desired valuation is at least $2e$ exactly when the nonnegative carry count vanishes. □

When b is a power of two, so are $\mathcal{B}, N_0, N_1, N$. On the range in (5), the low and high blocks separate the no-carry test into

$$\tau_2(g, M_c) = 0, \qquad \tau_2(J_c, (\mathcal{B}-2)T_*) = 0.$$

The first says exactly $g = \sum x_r \mathcal{B}^{w_r}$ with $0 \le x_r < b$: the low mask leaves those digit slots and those low bits free, and masks every other bit. At such a packed word, Theorem 2.1 gives target coefficients $2p_j(\mathbf{x})$. All coefficients of $C_{\mathbf{x}}\mathcal{D}$ have magnitude at most

$$M(1+\textstyle\sum x_r)^2 \le M(32b)^2 \le \mathcal{B}/4.$$

Thus the geometric offset in J_c gives, after its common binary factor 2 is removed, digits strictly between 0 and $\mathcal{B}$, with the target digit equal to $\mathcal{B}/2 + 2p_j(\mathbf{x})$. Its disjointness from $\mathcal{B}/2 - 1$ forces precisely $p_j(\mathbf{x}) = 0$.

It follows that a power-of-two b, the bound $0 < g < 4b\mathcal{B}^{200}$, and $N^2 \mid \binom{2R}{R}$ imply a common natural parent zero. Conversely, from a common parent zero choose an arbitrarily large power of two $b \ge b_*$ exceeding all its coordinates and pack them into g. The parent zero is nonzero, so $g > 0$; moreover $g < (b-1)\mathcal{B}^{201}/(\mathcal{B}-1) < b\mathcal{B}^{200}$. Both masks pass and $32g < G_c$. This is the stronger bound needed for the completeness direction of the ratio bridge.

3.3 The numerical Jones bridge

Continue with the following polynomial registers:

$$\begin{aligned}
U &= N^2 w, & Y &= N^2 s,\\
M_0 &= RY, & P &= 2M_0^2 U,\\
K &= R+1+h(P-1), & A &= M_0(U+1),\\
B_1 &= 2R+1, & W_1 &= bw,\\
C &= B_1 + W_1 + \phi, & J &= 4A-5,\\
Q &= 3W_1 C - 2(W_1^2-1), & D &= (A^2-1)C^2+1,\\
E &= iJDC^2, & F &= (A^2-1)E^2+1,\\
G &= A + F(CD+1-A), & H &= B_1 + jC,\\
I &= (G^2-1)H^2+1, & a_0 &= (P^2-1)K^2+1,\\
a_1 &= DFI, & u &= JF.
\end{aligned}$$

$$\begin{aligned}
v &= QF + J(H-C),\\
m &= K^2 G_c - 4(C-KY)^2 G_c - 8gK^2.
\end{aligned}$$

The numerical ratio lemmas used in the inherited construction state: given $R, N \ge 8$, $b \le R, N$, $b, G_c > 0$, the five positive bridge witnesses and $\phi \ge 0$, the conditions

$$a_0, a_1 \text{ are integer squares}, \quad u \mid v, \quad m > 0 \tag{6}$$

imply b is a power of two, $N^2 \mid \binom{2R}{R}$ and $8g < G_c$. Conversely the power-of-two and binomial conditions, with $32g < G_c$, admit witnesses satisfying (6). These are the repository's `Jones1982.Nine.ratio_sound` and `ratio_complete` contracts [9, 7]. All their numerical source hypotheses were established above. The old positional powers 5^j are not a hypothesis of these numerical lemmas; the new source coding is proved separately.

3.4 A single equation with no hidden quantified coordinates

Set

$$V_0 = PK, \qquad V_1 = (AC)(AE)(GH), \quad M_d = Q + H,$$
$$\Omega = um, \qquad \Theta = u(n + M_d) + v - \Omega(2M_d + V_0V_1).$$

The equation is

$$\boxed{F_9 = \left(\Theta^2 - \Omega^2(a_0 + V_0^2 a_1)\right)^2 - 4\Omega^4 V_0^2 a_0 a_1 = 0.} \tag{7}$$

This is a literal polynomial after the displayed substitutions. The final primary domain is nine positive integers: replace ϵ, ϕ, n in these formulas by $\epsilon - 1, \phi - 1, n - 1$. The other six coordinates remain positive. For the nine-natural version instead replace g, h, s, w, i, j by their successors and leave ϵ, ϕ, n nonnegative.

For completeness we recall why the inherited finalizer introduces no exceptional solutions. On every outer assignment all needed radicands are positive, and

$$0 < a_0 < V_0^2, \qquad 0 < a_1 < V_1^2, \qquad 0 < v/u < Q + H - 1 = M_d - 1.$$

The first two assertions follow from $V_0^2 - a_0 = K^2 - 1$ and $(AC)^2 - D = C^2 - 1$, $(AE)^2 - F = E^2 - 1$, $(GH)^2 - I = H^2 - 1$; the quotient estimate follows from $v/u = Q/J + (H - C)/F$ and $H > C$. If $m = 0$, then $\Omega = 0$ while $\Theta/u = n + M_d + v/u > 0$, so (7) cannot hold. Otherwise its four-factor identity makes $\Theta/\Omega = r_0 + V_0 r_1$ with $r_j^2 = a_j$. This rational sum is nonzero, for otherwise $|r_1| < 1$ would contradict $a_1 \in \mathbb{Z}_{>0}$. The identity $2(\Theta/\Omega)r_0 = (\Theta/\Omega)^2 + a_0 - V_0^2 a_1$ makes both roots rational and therefore integral. The strict majorants imply

$$Z = 2M_d + V_0V_1 + r_0 + V_0 r_1 \geq 2M_d + 1.$$

The vanishing factor becomes $n + M_d + v/u = mZ$, forcing $m > 0$ and $u \mid v$. Conversely choose integral roots under (6) and put $n = mZ - M_d - v/u$. This is an integer at least 2, so a factor vanishes. The source and ratio bridge now prove (1) for the stated domain.

3.5 Exact degree and the paid evaluator

Let $t_0 = \deg N = 3(8821 + 3 \cdot 200 + 2) = 28269$. The geometric offset uniquely dominates J_c, giving $\deg R = 3t_0$. The only large cancellation needed in the finalizer is resolved by the exact identity

$$V_1^2 - a_1 = (C^2 - 1)(AE)^2(GH)^2 + D(E^2 - 1)(GH)^2 + DF(H^2 - 1).$$

Writing $E_0 = u(n + M_d) + v - 2\Omega M_d$ gives identically

$$\Theta^2 - \Omega^2(a_0 + V_0^2 a_1) = E_0^2 - 2E_0\Omega V_0V_1 + \Omega^2 V_0^2(V_1^2 - a_1) - \Omega^2 a_0.$$

Degree propagation through this cancellation-resolved expression yields

$$\deg F_9 = 1152t_0 + 12W + 280 = 32\,568\,568.$$

On the diagonal where all nine coordinates have highest part T, its leading coefficient is $2^{2050} \neq 0$. Thus this is exact total degree, not just a syntactic upper bound. The affine domain changes preserve it. The primary graph has 652 live binary operations for the positive domain and 655 for the natural domain. The larger equal-polynomial degree-certificate graphs are separate verification artifacts and are not the claimed short evaluators.

4 The ten-integer companion

This section gives the complete evaluator over $\mathbb{Z}^{10}$. It uses the *unshifted* transport row in the table of Section 2, the same support-isolating labels, and the existing package-40/Sun arithmetic bridge. The two constructions therefore use slightly different parent coefficient records, despite having the same supported pairs. The ten unrestricted coordinates are exactly

$$f, g, h, k, l, w, x, y, z, m.$$

Every other symbol introduced in this section is a polynomial abbreviation, not an additional witness.

4.1 Fixed source and the required range contract

Let $\mathcal{D}$ be the factorial-weighted kernel of the unshifted parent, with the targets $E_j = 400 + 401(j-1)$. Thus its degree is 8821 and its leading coefficient is 2. Keep the whole package-31 tuple and put

$$H_0 = 8(1 + \beta + K_0 + M_C + M_F)^2, \qquad \beta_S = 2^{2140030}.$$

The ordinary quadratic coefficients have magnitude at most H_0, and the factorial-weighted ones have magnitude at most $2H_0$. For example, their port degree is at most two and their integer coefficient ℓ^1 norm is at most eight. The supplied coefficient audit verifies these statements exactly. The port bit lengths give

$$H_0 < 2^{2140017}, \qquad \beta_S > 4 \cdot 33^2 H_0.$$

The actual integers and all sixteen live nonliteral coefficient leaves are materialized in the archive. These coefficient choices are fixed; they do not add unknowns to the equation.

Define the source registers by

$$\begin{aligned}
b &= 1 + 3f, & \mathcal{B} &= \beta_S b^2, & \ell &= \sum_{r=1}^{31} \mathcal{B}^{w_r},\\
N_0 &= \mathcal{B}^{201}, & N_1 &= 4\mathcal{B}^{9222}, & N &= N_0 N_1 = 4\mathcal{B}^{9423},\\
M_c &= N_0 - 1 - (b-1)\ell, & T_* &= \sum_{r=1}^{22} \mathcal{B}^{E_r},\\
T_c &= M_c + (\mathcal{B} - 2)T_* N_0,\\
J_c &= 2(1+g)^2 \mathcal{D}(\mathcal{B}) + \mathcal{B} \sum_{r=0}^{9221} \mathcal{B}^r,\\
S_c &= g + J_c N_0,\\
R_c &= (S_c + T_c + 1)N + T_c + 1,\\
X &= (N-1)R_c, & Y &= N^2, & \mathcal{C} &= b\mathcal{B}^{200}.
\end{aligned}$$

In particular, the upper-range polynomial has the exact fixed form

$$\mathcal{C} = 2^{428006000} b^{401}. \tag{8}$$

Its power-of-two exponent is positive and even, and its exponent of b is positive and odd. All exponents in the source are fixed nonnegative integers.

Proposition 4.1 (Source contract for the ten-integer bridge)**.** *The displayed source has the following properties.*

(i) *From a common natural zero of the unshifted parent one can choose arbitrarily large $f > 0$, with b both a square and a power of two, and an integer g satisfying*

$$b \le g < \mathcal{C}, \qquad Y \mid \binom{2X}{X}.$$

(ii) *If $f \neq 0$, b is both a square and a power of two, $0 \le g < 2\mathcal{C}$, and $Y \mid \binom{2X}{X}$, then the parent has a common natural zero.*

(iii) *If $f \neq 0$, b is a square and $0 \le g < 2\mathcal{C}$, without yet assuming a power-of-two base, then*

$$0 \le S_c, T_c < N, \qquad 3 \mid X, \qquad X > 3b, \qquad Y > \max\{b, 256\}.$$

Proof. Under the hypotheses of (iii), $b = 1 + 3f$ forces $f > 0$ and $b \ge 4$. The positive leading coefficient and the bound $2H_0$ on the other kernel coefficients imply $\mathcal{D}(\mathcal{B}) > 0$. Integrality and $g < 2b\mathcal{B}^{200}$ give $1 + g \le 2b\mathcal{B}^{200}$ and $g < N_0$. Since $16H_0b^2 \le \mathcal{B} - 2$, the kernel estimate yields

$$\begin{aligned} 0 < J_c &\le (16H_0b^2 + \mathcal{B}) \sum_{r=0}^{9221} \mathcal{B}^r \\ &< 2\mathcal{B}^{9222} < N_1. \end{aligned}$$

The mask satisfies $0 \le M_c < N_0$, while $(\mathcal{B} - 2)T_* < \mathcal{B}^{8822} < N_1$. Hence both packed words lie in $[0, N)$. The exponent in β_S is even, so $\beta_S \equiv b \equiv \mathcal{B} \equiv 1 \pmod 3$ and $N \equiv 1 \pmod 3$. Therefore $3 \mid X$. Also $R_c \ge N + 1 > \mathcal{B} > b$ and $N \ge 4$, giving $X > 3b$; the fixed base makes $Y > \max\{b, 256\}$ immediate.

If b is a power of two, the words $\mathcal{B}, N_0, N$ are powers of two. Lemma 3.1 identifies the binomial divisibility with the joint no-carry test of S_c, T_c. Its low block forces $g = \sum_r n_r \mathcal{B}^{w_r}$ with $0 \le n_r < b$. Theorem 2.1 recovers the twenty-two residuals at the target cells. Every coefficient of the square times the kernel has absolute value at most

$$2H_0(1 + \textstyle\sum_r n_r)^2 < \mathcal{B}/2,$$

so the geometric offset and target masks test precisely whether all residuals vanish. This proves (ii).

Conversely, take a parent zero and choose $b = 2^{2q}$ as large as needed, with b exceeding all its coordinates. Then $f = (b - 1)/3$ is an arbitrarily large positive integer. Pack the zero into g. It is not the all-zero tuple, and all selected weights are positive, so $g \ge \mathcal{B} \ge b$. The geometric bound gives $g < b\mathcal{B}^{200} = \mathcal{C}$. The same masks pass, and the binomial divisibility follows. This proves (i) and completes the contract. □

4.2 The complete polynomial evaluator

Continue with the following substitutions:

$$\begin{aligned} L &= lY, & U &= 2LX,\\ V &= 4gwY, & W_b &= bw,\\ A &= U(V+1), & B_1 &= 2X+1,\\ C &= B_1 + (A-2)h, & P_n &= U^2V,\\ K &= X+1+k(P_n-2), & S &= 2A-5,\\ T &= 3W_bC - 2(W_b^2-1), & R_2 &= (P_n^2-4)K^2+4. \end{aligned}$$

$$\begin{aligned} \Delta_b &= 8\mathcal{C}^3gK^2 - g^2(32(C-KL)^2\mathcal{C}^3 + g^2K^2),\\ O &= f^2l^2\Delta_b, \qquad \xi = zx. \end{aligned}$$

The modified Lucas-certificate registers are

$$\begin{aligned} D &= (A^2-4)C^2+4, & E &= C^2D\xi,\\ F &= 4(A^2-4)E^2+1, & G_* &= 2(A+2)(A-2)^2E^2-1,\\ \omega &= 16E, & H_* &= C + (B_1-C)F + \omega Fy,\\ I_* &= (G_*^2-1)H_*^2+1. \end{aligned}$$

Define the divisibility merger, radicands and separating weights by

$$\begin{aligned} \Sigma &= \xi S^5, & r &= T + Sy, & &\\ \Theta &= (r^3 - rS^2 - S^3)(r^2+23S^2), & & & &\\ r_0 &= b(1+(3O-4)b^2z^2), & r_1 &= R_2, & r_2 &= DFI_*,\\ v_0 &= 1, & v_1 &= r_0, & v_2 &= r_0(P_n^2+K^2+3). \end{aligned}$$

Finally put

$$\begin{aligned} Z_n &= \Sigma m - \Theta, & a_i &= \Sigma^2 v_i^2 r_i \quad (0 \le i \le 2),\\ V_n &= Z_n^2 + a_2 - a_0 - a_1, & & \\ W_n &= V_n^2 + 4Z_n^2a_2 - 4a_0a_1. & & \end{aligned}$$

The requested unrestricted-integer equation is

$$\boxed{F_{10} = W_n^2 - 16V_n^2Z_n^2a_2 = 0.} \tag{9}$$

Together with the fixed coefficient records and the source above, this specifies every arithmetic operation needed to define one polynomial in the ten displayed integer unknowns. There are no divisibility predicates, roots, variable exponents or hidden existential parameters in its definition.

4.3 The imported parameterized arithmetic bridge

We state the precise inherited interface rather than silently applying a theorem whose literal statement uses a different source polynomial. Package 40 reads the proofs of Sun's Propositions 6.1–6.2 parametrically: the source enters through Proposition 4.1, $b = 1+3f$, and a bound of the form $\mathcal{C} = 2^c b^d$ with fixed positive even c and positive odd d. Its size and congruence

hypotheses are exactly those proved in part (iii). The new constants $c = 428006000$ and $d = 401$ satisfy that interface. The polynomial tail and the arithmetic witnesses have not changed [5, 10, 11].

For clarity, let $u_0(A) = 0$, $u_1(A) = 1$ and $u_{q+1}(A) = Au_q(A) - u_{q-1}(A)$, extended in the usual way to integer indices. The imported bridge and interpolation statements are:

1. A parent zero admits f, g, h, k, l, w with $A > 2$ even, $B_1 > 0$ odd, b a square, $b \le g < \mathcal{C}$, $C = u_{B_1}(A)$, R_2 a square, $S \mid T$, and $O > 0$.

2. For the displayed substitutions, the conditions

$$b \text{ square}, \quad O > 0, \quad \xi \neq 0, \quad DFI_* \text{ square}, \quad R_2 \text{ square}, \quad S \mid T$$

   imply a parent zero.

3. If $A > 2$ is even, $B_1 > 0$ is odd and $C = u_{B_1}(A)$, then for every positive integer d_0 there is $\xi > 0$ divisible by d_0 such that each residue class modulo ξ contains a y for which DFI_* is a square.

The third statement is Sun's full residue-class interpolation theorem. It is needed because the divisibility merger below prescribes a residue class after choosing ξ. We retain this theorem as a cited dependency; we do not claim a fresh proof of the Lucas results or of the general ten-integer theorem.

The reverse direction does not assume a power-of-two base prematurely. Once b is a square and $O > 0$, the displayed formula for O forces $f, l \neq 0$, $\Delta_b > 0$, $g > 0$ and $K \neq 0$. It then gives

$$0 < g < 2\mathcal{C}, \qquad 4(C - KL)^2 < K^2.$$

Part (iii) of Proposition 4.1 is therefore available before the Lucas step. The inherited reverse argument recovers

$$C = u_{B_1}(A), \quad K = u_{X+1}(P_n), \quad W_b = 2^{B_1}, \quad Y \mid \binom{2X}{X}.$$

Since $W_b = bw$ and $b > 0$, it follows that b is a power of two. Only at this point is part (ii) applied. This order of reasoning is why the broad pre-decoding range contract was proved separately.

4.4 Why the finalizer introduces no extra solutions

Here we recall the short algebraic part of the inherited finalizer, including its nonvanishing and choice-order conditions. The integer $S = 2A - 5$ is always odd and nonzero. The polynomial

$$\pi(v) = (v^3 - v - 1)(v^2 + 23)$$

has no rational root and has a root modulo every positive integer. For the latter assertion, at an odd prime other than 23 either the cubic has a simple root or it is irreducible; in the irreducible case Frobenius is a three-cycle, so its discriminant -23 is a square and the quadratic has a simple root. At 23 the cubic has the simple root 3. These roots lift to odd prime powers. At 2, the fact $-23 \equiv 1 \pmod 8$ supplies a root in the 2-adic integers. The Chinese remainder theorem finishes the claim. The rational-root test for the cubic and positivity of the quadratic prove nonvanishing over $\mathbb{Q}$.

If $S \mid T$, write $T = Sq$. For any $\xi \neq 0$ choose a root c_0 of π modulo $|\xi|$ and prescribe $y \equiv c_0 - q \pmod{\xi}$. Then $\Sigma \mid \Theta$. Conversely, $S^5 \mid \Theta$ forces $S \mid T$: if $\nu_p(S) > \nu_p(T)$, the two homogeneous factors of Θ have valuations $3\nu_p(T)$ and $2\nu_p(T)$, contradicting $S^5 \mid \Theta$. Also $\Theta \neq 0$, since otherwise $\pi((T + Sy)/S) = 0$.

Suppose $z \neq 0$ and r_0 is a square. The factors b and $1 + (3O - 4)b^2z^2$ are coprime. The second cannot be zero: that would force $|b| = |z| = 1$, hence $b = 1$, $f = O = 0$, contradicting its vanishing. Unique factorization makes b a signed square; $b \equiv 1 \pmod 3$ excludes the negative sign. The other factor is then a square as well, which forces $O > 0$ because for $O \leq 0$ it is at most $1 - 4(bz)^2 < 0$.

For the converse, the positive bridge gives $O \geq g \geq b \geq 4$: Δ_b is a positive multiple of g, and f, l are nonzero. Thus $d = 3O - 4$ is positive and is 2 modulo 3, so it is nonsquare. A positive Pell unit for d, raised to a suitable finite order modulo b, supplies $u^2 - dv^2 = 1$ with $v > 0$ and $b \mid v$. Choose $z = v/b > 0$. Then r_0 is a square. The interpolation theorem can now choose $\xi > 0$ divisible by this already fixed z; put $x = \xi/z$. Only after this modulus exists do we prescribe the residue class of y needed for $\Sigma \mid \Theta$.

For every integer assignment with $z \neq 0$, one has $|r_0| > 1$. Indeed $r_0 = 0, \pm 1$ would force $|b| = 1$, hence $b = 1$ and $f = O = 0$, whereas $r_0 = 1 - 4z^2 \leq -3$. Also

$$\sqrt{|r_1|} \leq P_n^2 + K^2 + 2,$$

as follows by subtracting $(P_n^2 + 4)K^2 + 4$ from $(P_n^2 + K^2 + 2)^2$. Consequently

$$|v_1| > \sqrt{|r_0|}, \qquad |v_2| > \sqrt{|r_0|} + |v_1|\sqrt{|r_1|}.$$

If a signed sum of these three weighted roots is rational, every radicand is an integer square. To see this, in the multiquadratic extension take the largest-index nonzero root term changed by an automorphism. Its change has greater absolute value than all preceding changes combined, a contradiction. This argument works with complex absolute values for negative radicands and does not assume independent square classes. Rational square roots of integers are integral.

The final polynomial has the exact identity

$$F_{10} = \prod_{\varepsilon \in \{\pm 1\}^3} \left(Z_n + \varepsilon_0\sqrt{a_0} + \varepsilon_1\sqrt{a_1} + \varepsilon_2\sqrt{a_2}\right).$$

The radicals cancel; successive conjugate multiplication gives precisely (9). If $\Sigma = 0$, then all $a_i = 0$ and $F_{10} = \Theta^8 \neq 0$. At any zero, therefore, $\xi = zx \neq 0$. Divide a vanishing factor by Σ. Separation makes all r_i squares and the resulting signed sum integral, so $\Sigma \mid \Theta$. The two mergers give $S \mid T$, b square and $O > 0$; the reverse bridge then yields a parent zero.

Conversely, a parent zero gives the positive bridge witnesses. Choose z, ξ, x and y in the order just described, obtaining all three square radicands and $\Sigma \mid \Theta$. An integer m can then make Z_n equal to the negative of Σ times any signed sum of the weighted integer roots. One norm factor vanishes. This proves both implications for the ten-integer equation and hence (1), using the explicitly stated inherited compiler and arithmetic contracts.

4.5 Exact degree and evaluator size

Put $u = 9221 + 200 + 2 = 9423$. The geometric offset uniquely dominates J_c, and the source registers have degrees

register	N	S_c	T_c	X	Y	$\mathcal{C}$
degree	$2u$	$2u$	$2u - 800$	$6u$	$4u$	401.

In particular $d_X = 56538$, $d_Y = 37692$ and $d_C = 401$. Propagation through the retained tail gives

$$\begin{aligned}\deg Z_n &= 5d_X + 10d_Y + 30 = 659640,\\ \deg a_0 &= 14d_X + 28d_Y + 3d_C + 57 = 1848168,\\ \deg a_1 &= 26d_X + 48d_Y + 6d_C + 98 = 3281708,\\ \deg a_2 &= 114d_X + 224d_Y + 6d_C + 440 = 14891186.\end{aligned}$$

The last degree exceeds those of a_0, a_1, Z_n^2. Therefore the highest part of F_{10} is the fourth power of the highest part of a_2, giving

$$\deg F_{10} = 456d_X + 896d_Y + 24d_C + 1760 = 59\,564\,744.$$

The independent actual-coefficient certificate also gives

$$[T^{59564744}]F_{10}(T,\dots,T) \equiv 285435419 \pmod{1000000007},$$

which is nonzero. This is an exact leading-coefficient computation, not sampling of possible solutions.

The primary division-free graph has 215 additions, 25 subtractions and 413 multiplications, totaling **653 operations**. Shared powers, row-wise quadratic Horner evaluation and divide-and-conquer geometric sums account for its size. Every binary gate is charged under the common convention. These are exact construction metrics and do not assert that no shorter ten-integer evaluator exists.

5 Eleven positive variables: combining quotient absorption with a smaller norm

Package 38 removes the Pell index quotient h by absorbing its divisibility condition into an existing carrier. Package 44 removes an inner radical separator, but its displayed root estimate uses $k > E$, which was obtained from that same quotient equation. Combining the two constructions therefore requires a new estimate, not just substituting one formula into the other. The estimate below compares the coupled-root gap with the actual relative error. It works before h has been reconstructed and preserves the smaller weight $K = IT + 1$.

5.1 The two explicit equations

The supplied variables are precisely

$$J, F, \alpha, z, i, T, s, w, \eta, \zeta, t \in \mathbb{Z}_{>0}.$$

Use the complete fixed tuple (β, K_0, M_C, M_F) specified in the coefficient appendix. The following names are polynomial abbreviations, not additional unknowns:

$$\begin{aligned}
q &= \beta J + 1, & s_0 &= s + 3q, & C_w &= q - F - \alpha,\\
X &= wq, & Y &= s_0 q^3, & E &= XY,\\
k &= \eta + \zeta, & c &= kY + \eta, & a &= E + Y,\\
H &= 4a + 3, & \Delta &= a^2 + H = (a+2)^2 - 1, & \ell &= EkY,\\
A &= 1 + \ell(\ell + k), & G &= 1 + \Delta c^2, & b_0 &= X + ac,\\
I &= ic^2, & D &= 1 + \Delta I^2, & Q &= \Delta(D-1) = (\Delta I)^2,\\
d &= Q - 1, & v &= cT.
\end{aligned}$$

Define also

$$\begin{aligned} R &= ((q-1)(q-F)+\alpha)(q^2-1)+(M_C+qM_F)J,\\ N_t &= (K_0+w)C_w+(q-F)-z(q-1),\\ b &= c+RD, \qquad e = Q(Dv^2+b^2)-1, \qquad B_a = 2Qvb. \end{aligned}$$

Choose one of two carrier corrections,

$$\delta_{\mathrm{f}} = (a+2)c-\ell \quad \text{or} \quad \delta_{\mathrm{s}} = ac-\ell.$$

The subscripts denote the full and short carriers. For either choice put

$$\begin{aligned} K &= IT+1, \qquad S = EH, \qquad L = E(1+HK),\\ W &= Eb_0+H\big(E(t+b+\delta)+k-R-1\big). \end{aligned} \tag{10}$$

The computed carrier weight K is distinct from the fixed compiler coefficient K_0.

Define the integer polynomial

$$\mathcal{N} = \prod_{\substack{g=\pm\sqrt{G}\\ \tau=\pm\sqrt{A}\\ f=\pm\sqrt{D}}} \Big[d(W-Lg-S\tau)^2 - S^2e + S^2B_af\Big]. \tag{11}$$

The signs in this display specify a formal eight-factor product. Each root appears through an even polynomial after the product is expanded, so $\mathcal{N}$ has integer coefficients. The notation does not add quantified variables or assert that independent Galois automorphisms of the three radicals exist. The requested scalar equation is

$$\boxed{P_{11} = N_t(1+\mathcal{N})-1 = 0.} \tag{12}$$

For an entirely division-free evaluation of (11), set

$$\begin{aligned} n_d &= dS^2A, & H_P &= S^2QDv^2, & H_U &= S^2Qb^2,\\ H_e &= H_P+H_U-S^2, & j_0 &= 4H_PH_U,\\ \mathcal{T} &= 2d(W^2+L^2G), & \mathcal{K} &= [d(W^2-L^2G)]^2,\\ A_1 &= 2(n_d+H_e), & c_1 &= (n_d-H_e)^2-j_0,\\ a_1 &= \mathcal{T}-A_1, & b_1 &= c_1-\mathcal{K},\\ v_1 &= 16n_dj_0-a_1(a_1\mathcal{T}+2b_1), & u_1 &= b_1^2-a_1^2\mathcal{K}. \end{aligned}$$

Then

$$\mathcal{N} = \mathcal{K}v_1^2 - u_1(\mathcal{T}v_1-u_1). \tag{13}$$

One verifies this identity by eliminating τ, f first and reducing the resulting quartic in $d(W-Lg)^2$ modulo its quadratic relation. It is a polynomial identity, including at square or coincident radicands, not an identity restricted to successful witnesses.

Theorem 5.1. *For each choice of δ, the polynomial (12) has a zero in $\mathbb{Z}_{>0}^{11}$ if and only if the normalized blank-K215 compiler has a solution. Thus its absence of positive zeros is provably equivalent in ZFC to* $\mathrm{Con}(\mathsf{ZFC})$*, through the inherited compiler and machine/refutation calibration. The full carrier has exact degree* 990 *and a* 117*-operation evaluator; the short carrier has exact degree* 994 *and a* 115*-operation evaluator.*

5.2 Extracting the unit and obtaining the outer bounds

Put $x_0 = d(W^2 + L^2G)$. The scalar evaluator satisfies

$$\mathcal{T} = 2x_0, \qquad \mathcal{K} = x_0^2 - 4d^2W^2L^2G,$$

and consequently

$$\mathcal{N} \equiv (u_1 - x_0v_1)^2 \pmod 4. \tag{14}$$

At a zero of P_{11} both N_t and $1+\mathcal{N}$ are integer units. The possibility $1+\mathcal{N} = -1$ would give $\mathcal{N} = -2$, contradicting (14). Therefore

$$\mathcal{N} = 0, \qquad N_t = 1.$$

The authentic coefficient interface now gives

$$\begin{gathered} q \geq 16, \quad C_w > 0, \quad 3q < R < q^4 - 2, \\ X \geq q, \quad Y = (s+3q)q^3 > 3q^4 > 3R, \quad E > R + 1, \\ E \geq 65536, \quad k \geq 2, \quad kY < c < k(Y+1) < 2kY, \\ I \geq c^2, \quad c \geq 2, \quad a + 2 < 2E. \end{gathered} \tag{15}$$

For example, $C_w \leq 0$ would give $N_t \leq 1 - F - (z-1)(q-1) \leq 0$, so the boundary $C_w = 0$ is excluded here by the already recovered transport sign. No assertion $k > E$, $c > EY$, or $R < k$ is used before the omitted quotient is recovered.

5.3 A relative-error estimate replaces the lost quotient bound

For estimates only, let $g = \sqrt{G}$, $f = \sqrt{D}$ and $\tau = \sqrt{A}$ denote the positive real roots. From (15),

$$\begin{gathered} g < (a+2)c < 2Ec, \qquad f < (a+2)I < 2EI, \\ 1 < A < \Delta I = \sqrt{Q}, \qquad g > \tau. \end{gathered} \tag{16}$$

For the radicand comparison, $\Delta c^2 - A$ exceeds $k^2[2EY^3 + Y^4 - EY] - 1 > 0$; also $I \geq c^2$. These estimates do not require the index equation.

Lemma 5.2 (The sufficient coupled-root gap)**.** *Under* (15)*,*

$$\frac{b + cf}{d} < Ig - cf.$$

Every coherent auxiliary root, including a complex root, satisfying

$$dy^2 = Q(vf' - b)^2 - 1 = e - B_af', \qquad f' = \pm f,$$

therefore obeys

$$|y| < b + ITg, \qquad |\tau' + y| < b + (IT+1)g \quad (\tau' = \pm\tau).$$

Proof. The exact identity $(Ig)^2 - (cf)^2 = I^2 - c^2$ first gives $Ig > cf$. Since $I \geq c^2$ and $c \geq 2$, its right side is greater than $I^2/2$. Thus

$$Ig - cf > \frac{I^2}{4Ig} > \frac{I}{8Ec} \geq \frac{c}{8E} > \frac{Y}{4E}.$$

On the other hand $c < I^2$, $D < 4E^2I^2$, $R < E$ and $cf < 2EI^2$ give

$$b < 5RE^2I^2 < 5E^3I^2, \qquad b + cf < 6E^3I^2, \qquad d > \tfrac{1}{2}E^4I^2.$$

Consequently

$$\frac{b+cf}{d} < \frac{12}{E} < \frac{Y}{4E} < Ig - cf,$$

because $Y > 48$. The difference $Ig - cf$ need not exceed one: comparison with this smaller error is the new point.

Set $Z = b + Tcf > 1$. Taking complex absolute values and using $Q = d + 1$,

$$|y|^2 \le \frac{QZ^2 + 1}{d} < \big[Z(1 + 1/d)\big]^2.$$

The last strict difference is $(QZ^2 - d)/d^2 > 0$. Since $T \ge 1$,

$$Z/d \le T(b + cf)/d < T(Ig - cf).$$

Hence $|y| < Z + Z/d < b + ITg$. Finally $\tau < g$ proves the stated bound for $|\tau' + y|$. □

5.4 Recovering the radicals and the absorbed divisibility

Eliminating the two inner radicals gives the rational-coefficient polynomial

$$C(x) = d(x^2 + A) - e,$$
$$\mathcal{F}(x) = [C(x)^2 + DB_a^2 - 4d^2x^2A]^2 - 4C(x)^2DB_a^2.$$

At $\mathcal{N} = 0$ some branch gives $x = (W - Lg')/S = \tau' + y$ and $\mathcal{F}(x) = 0$. If $g' = -g$, then

$$x = b + \left[t + \frac{k - R - 1}{E}\right] + \delta + \frac{b_0}{H} + (K + 1/H)g > b + Kg.$$

The bracket is positive because $t \ge 1$, $k \ge 2$ and $E > R+1$; both carrier corrections are positive. This contradicts Lemma 5.2. If G were nonsquare, the actual quadratic conjugation in $\mathbb{Q}(\sqrt{G})$ would carry a root at one main sign to the excluded sign. Therefore g is a positive integer and x is rational.

The exceptional inner branches can be excluded without a separator. If $C(x) = 0$ at a root, substitution gives the integer identity

$$d^2A^2 - deA + Q^2v^2b^2D = 0.$$

Modulo Q, both d and e are -1, so $Q \mid A(A-1)$. This is impossible because $1 < A < \sqrt{Q}$. Likewise $\mathcal{F}(0) = 0$ would force $Q \mid (A-1)^2$, also impossible. Thus both denominators in the following reconstruction are nonzero:

$$U_x = C(x)^2 + DB_a^2 - 4d^2x^2A,$$
$$f' = -\frac{U_x}{2C(x)B_a}, \qquad \tau' = \frac{C(x) + B_af'}{2dx}, \qquad y = x - \tau'.$$

The equation $\mathcal{F}(x) = 0$ gives $(f')^2 = D$, $(\tau')^2 = A$ and the auxiliary equation. The rational roots f', τ' are integers.

We use the inherited rational Pell saturation lemma: for integers $m \ge 2, V$, a rational solution of $m^2V^2 - (m^2-1)y^2 = 1$ is integral. Briefly, writing $y = u/v$ in lowest terms gives $v^2 \mid m^2 - 1$. In the integer-coordinate Pell group of discriminant $(m^2-1)/v^2$, the solutions (m, v) and $(m|V|, |u|)$ have indices j, l. Divisibility of their abscissas forces l to be an odd multiple of j: reduce the index modulo $2j$, reflect into $[0, j]$, and use strict growth of the abscissas. The ordinate divisibility then gives $v \mid u$, hence $v = 1$. Apply this lemma with $m = \Delta I$ and $V = vf' - b$. It makes y and consequently x integral. Reducing $W = Lg + Sx$ modulo E yields

$$E \mid H(k - R - 1). \tag{17}$$

The factor H has not yet been cancelled.

5.5 The short-range argument restores the full index

Let χ_{a+2}, ψ_{a+2} be the Pell sequences. The recovered main and normalized strong equations give indices $p, m \geq 1$ with

$$g = \chi_{a+2}(p), \quad c = \psi_{a+2}(p), \quad |f'| = \chi_{a+2}(m), \quad I = \psi_{a+2}(m).$$

Since $c^2 \mid I$, strong divisibility and composition first give $p \mid m$ and then $c \mid m/p$. In particular $c \mid m$ and $m \geq pc > 2p$. The auxiliary odd-index identities give a further index j_0 such that

$$j_0 \equiv \pm p \pmod{m}, \qquad j_0 \equiv \pm R \pmod{c}.$$

These are the normalized signed-quotient identities used in package 38: they follow from $V \equiv -c \pmod{|f'|}$, $V \equiv -R \pmod{c}$, the odd Pell quotient polynomial, and signed index reduction. They allow either sign of f' or V. As $c > 2Y > 6R$ and $c > 2p$, they imply $p = R$ before any use of h. For the latter bound, $c > 1$ gives $p \geq 2$, and the Pell recurrence gives $c \geq 5^{p-1} > 2p$.

The first equation classifies

$$|\tau'| = \chi_{2EY+1}(n), \qquad k = 2\psi_{2EY+1}(n), \qquad 1 \leq n < p = R.$$

The last inequality follows by parameter monotonicity from $2EY + 1 > a + 2$ and $c > k$. Put $\mu = E/\gcd(E, H)$. Because $\gcd(Y, H) = \gcd(Y, 3) \leq 3$,

$$\mu \geq Y/3 > R.$$

Equation (17) gives $k \equiv R+1 \pmod{\mu}$. The first Pell recurrence gives $k \equiv 2n \pmod{E}$. Hence $2n \equiv R+1 \pmod{\mu}$; since $|2n - R - 1| < R < \mu$, this means $2n = R+1$ exactly. It follows that

$$h = (k - R - 1)/E \in \mathbb{Z}_{>0}.$$

Indeed $R > 48$ gives $n \geq 25$, so strict Pell growth gives $k > 2n$.

Now divide the carrier identity by E and reduce modulo H. This gives $g \equiv b_0 \pmod{H}$, so $\gamma = (g - b_0)/H$ is an integer. It is positive: $n \geq 25$ implies $k \geq 4(2EY + 1) > X$, hence $c > X$, whereas $g > (a + 1)c$. We have recovered every equation of the normalized compiler. Taking absolute values of τ', y is harmless; the inherited signed-quotient interface permits the recovered auxiliary signs. This proves soundness using the identified compiler theorem, not by assuming its conclusion during radical recovery.

5.6 Completeness and exact complexity

For a blank halting run, choose canonical positive normalized compiler witnesses. Their estimates give $Y > 2^R$, $R > 3q$ and $q \geq 16$, so $Y/q^3 > 3q$ and $s = Y/q^3 - 3q$ is positive. For both corrections, $0 < \delta < g$: since $c < 2kY$, one has $\ell/c > E/2 > 2$, and $(a+2)c - \ell < ac < g$. Define

$$t = \gamma + Kg + \tau + y - b - \delta - h.$$

The auxiliary equation gives $y \geq |Tcf - b| \geq b - Tcf$. Consequently

$$t > \gamma + \tau - h + T(Ig - cf) > 0,$$

because $\tau > \ell > k/E > h$ and $Ig > cf$. Its defining identity gives $W = Lg + S(\tau + y)$, so a coherent factor in (11) vanishes and $N_t = 1$. This proves completeness.

The exact degrees can be transferred without assuming generic coefficients. Let $\mathcal{N}_{44}$ be package 44's corresponding full or short norm, before quotient absorption. In the localization at E, there is the exact identity

$$\mathcal{N} = E^{16}\mathcal{N}_{44}\left(s + 3q,\, t + \frac{k - R - 1}{E}\right). \tag{18}$$

Each of its eight factors is multiplied by E^2. The old norm has t-degree at most 16, so denominators clear and this is an ordinary polynomial identity. Its source exact degrees are 892 and 896 for the full and short carriers respectively [6]. The rational shift has degree $4 - 6 = -2$, strictly below $\deg t = 1$; any term using it loses at least three degrees. The highest part of $s \mapsto s + 3q$ is the invertible linear substitution $s \mapsto s + 3\beta J$, so it preserves the nonzero leading form. Since $\deg E = 6$, (18) gives exact norm degrees 988 and 992. Finally N_t has nonzero leading form $w(\beta J - F - \alpha) - z\beta J$, of degree two, giving exact scalar degrees 990 and 994.

The literal division-free DAGs charge all operations in these definitions and contain no dead gates. They give 117 operations for the full carrier and 115 for the short carrier. The degree proof, the charged gate counts, and the semantic equivalence are separate claims: the finite circuit audits support the first two, while the argument above and its explicit inherited Pell/compiler dependencies establish the third. No new end-to-end formal certificate or absolute minimality claim follows from this construction.

6 Fifteen positive variables at degree 44

The low-degree supplement is a polynomial in fifteen strictly positive integer variables of exact degree 44 evaluated by 76 binary operations: 41 multiplications, 19 additions, and 16 subtractions. This improves the earlier degree-44 evaluator's 78 operations by two. The root-gap substitutions themselves are inherited: historical package 12 used the auxiliary gap, and package 15 used both gaps. Package 10's Q_{56} used four grouped residuals, 74 operations, and degree 56. The change here is their combination with a different pairing of the norm, index, and transport factors. It is neither the first degree-44 construction nor an operation-count improvement over P_{67} [8, 2].

All four coefficient ports use package 31's complete tuple [3], $B = 2^{1064921}$. Write them as β, K_0, M_C, M_F, where $\beta = B - 1$ and $M_F = M_F^{\text{native}} + B - 1$ is the already shifted high mask. The tuple is specified in Appendix A. The original package 36 binding in the copied active evaluator is not retained; no identity with its fixed Lean polynomial is asserted.

The names in this section are local. This is a scalar presentation of the same ordinary strong-norm compiler interface as the parent in Section 2; every abbreviation is defined again below. The multiplier i is an ordinary multiplier, not a normalized multiplier. The variables are

$$J, F, \alpha, z, f, h, i, T, s, w, r, \eta, \zeta, u, \gamma \in \mathbb{Z}_{>0}.$$

The following quantities are polynomial abbreviations:

$$\begin{aligned}
q &= \beta J + 1, & C &= q - F - \alpha, & & \\
X &= wq, & Y &= sq^3, & E &= XY, \\
k &= \eta + \zeta, & c &= kY + \eta, & a &= E + Y, \\
H &= 4a + 3, & \Delta &= a^2 + H, & D &= X + ac + \gamma H, \\
L &= E(kY), & G &= (q-1)(q-F) + \alpha, & & \\
R &= G(q^2 - 1) + (M_C + qM_F)J, & & & & \\
V &= c(Tf - 1) - Rf^2, & A_0 &= \Delta(f^2 - 1), & I &= ic^2.
\end{aligned}$$

Define

$$\begin{aligned} N_0 &= r^2 + L(2r-k), & N_1 &= D^2 - \Delta c^2, \\ N_a &= (V+u)^2 - A_0 u(2V+u), & & \\ N_k &= k - hE - R, & N_t &= (K_0+w)C + (q-F) - z(q-1). \end{aligned}$$

The proposed single equation is

$$\boxed{(N_0N_k-1)^2 + (N_1N_t-1)^2 + (N_a-1)^2 + (I^2-A_0)^2 = 0.} \tag{19}$$

Its literal division-free evaluation is in `short/offset44.json`.

Soundness. A zero of (19) gives $N_0N_k = N_1N_t = N_a = 1$ and $I^2 = A_0$. Restore $\tau = L+r > 0$. The exact identity

$$(L+r)^2 - L(L+k) = r^2 + L(2r-k)$$

expresses N_0 as $\tau^2 - n(n+1)k^2$ for $n = XY^2 \geq 2$. The negative Pell equation with this consecutive-product discriminant has no solution with $k > 0$: multiplication by $2n+1-2\sqrt{n(n+1)}$ decreases the positive second coordinate of any putative norm-minus-one solution. Thus the unit N_0 is positive and $N_0 = N_k = 1$. Similarly $\Delta = (a+2)^2 - 1$ is 0 or 3 modulo 4, so N_1 cannot be -1; consequently $N_1 = N_t = 1$.

Put $y_0 = V + u$. The second exact identity is

$$A_0(V^2 - (V+u)^2) + (V+u)^2 = (V+u)^2 - A_0u(2V+u).$$

If $y_0 = 0$, the equation $N_a = 1$ would imply $I^2V^2 = 1$. But $I = ic^2 > 1$, so this is impossible. Hence $y = |y_0| > 0$. The restored positive coordinates solve the inherited ordinary strong system

$$\tau^2 - L(L+k) = 1, \qquad N_1 = 1, \qquad A_0(V^2-y^2) + y^2 = 1,$$
$$N_k = N_t = 1, \qquad I^2 = A_0.$$

The authenticated ordinary strong-rank and compiler theorems therefore give blank K215 halting [3]. This argument does not assume that the computed expression $V + u$ is positive away from zeros, and it does not divide i by Δ.

Completeness. Start from the canonical positive normalized witness supplied by a blank halting computation and use the inherited transfer $i_{\text{ordinary}} = \Delta i_{\text{normalized}}$. It yields the ordinary system above with $V > 1$, $y > 0$, and $A_0 > 1$. The strict inequality $V > 1$ is essential: it excludes the degenerate auxiliary root $V = y = 1$ whose gap would vanish. It follows from the inherited auxiliary positivity argument and the congruence $V \equiv -c \pmod f$ with $f > 2c > 2$. Set $r = \tau - L$ and $u = y - V$. The norm equations give

$$\tau^2 - L^2 = Lk + 1 > 0, \qquad (A_0 - 1)(y^2 - V^2) = V^2 - 1 > 0.$$

Therefore both new coordinates are strictly positive, and the two identities prove (19). Composing with the ordinary compiler and K215 calibration establishes, by ordinary mathematics formalizable in ZFC, that the absence of positive zeros is equivalent to Con(ZFC). No new proof-assistant certificate or checked object-language ZFC derivation is claimed.

Exact degree. The five factor degrees are respectively $12, 18, 22, 7, 2$; $I^2 - A_0$ has degree 22. For the main factor, setting $Z = X + \gamma H$ gives the identity

$$(ac + Z)^2 - (a^2 + H)c^2 = Z^2 + 2acZ - Hc^2,$$

which lowers its apparent degree 22 to 18. The four residual degrees in (19) are consequently $19, 20, 22, 22$. With $Q = \beta J$ and $k_0 = \eta + \zeta$, the degree-44 homogeneous part is

$$i^4 k_0^8 s^8 Q^{24} + 4k_0^2 w^4 s^6 T^2 f^6 u^2 Q^{22}.$$

The monomial $i^4\eta^8 s^8 J^{24}$ has nonzero coefficient β^{24}. Thus 44 is the exact degree. The standard-library audit independently checks the literal 76-gate schedule, all ports' liveness, the three polynomial identities, and this complete homogeneous part. Normal and optimized Python runs give the same ledger.

Historical comparison. The wave-2 pin `5b0c9ae9fd` contains package 10's `ordinary_blank_four_sos_74.json`, with the positive-domain proof in `sections/ordinary_section.tex`; package 12's `GAP_AUXILIARY_PROOF.md` and degree-44/78-operation evaluator; and package 15's `POSITIVE_GAP_SHEARS.md`, with the degree 83/71 product and degree 44/80 six-square version. These are like-for-like comparisons of positive-domain templates and paid binary operations, with all coefficient tuples treated coherently. The two-operation saving is a bounded repository comparison, not a global priority assertion.

7 Reproduction, evidence and remaining questions

7.1 Canonical objects in the archive

The mathematical definitions above are accompanied by machine-readable straight-line programs. Their variables, fixed coefficient bindings, ordered binary gates and output names make them exact polynomial specifications without requiring a fully expanded coefficient list. The principal files are as follows.

Object	Canonical file
Nine positive	`nine/generated/multitarget_nine_positive.json`
Nine natural	`nine/generated/multitarget_nine_natural.json`
Ten integer	`data/polynomial.json`
Eleven, degree 990	`eleven/generated/eleven_full_product.json`
Eleven, degree 994	`eleven/generated/eleven_short_product.json`
Fifteen, degree 44	`short/offset44.json`
Shared coefficients	`coefficients/COEFFICIENT_BINDING.json`
Extra packed leaves	`fixed/FIXED_LEAVES.json`
Complete audit driver	`verify_all.py`

Auxiliary systems, sum-of-squares variants and larger degree-certificate circuits are named separately. The current operation counts refer only to the canonical primary objects listed above.

7.2 What the finite audits establish

The driver runs mutually distinct checks of the relevant finite claims:

1. An independent reconstruction of the four native integers from the finite coefficient recipe and literal tiles, with byte-level hashes.

2. Exhaustive enumeration of all 528 padded pair sums, isolation of all 71 active pairs, and a deliberately invalid-label negative control.

3. Exact symbolic reconstruction of both quadratic parent systems and all 44 extraction identities, retaining the native ports as independent polynomial symbols. The transport shift is checked explicitly.

4. Exact reconstruction and hash checks of all 33 additional fixed leaves, with the coefficient bounds needed in the carry arguments.

5. Gate ordering, permitted operations, witness and coefficient bindings, complete liveness, and the claimed operation ledgers.

6. Exact algebraic identities used for norm cancellation and the short sum-of-squares form, together with independently evaluated full formulas. Finite modular evaluations supplement these identities.

7. Degree upper bounds from cancellation-resolved expressions and nonzero leading forms. The eleven-variable audit computes complete polynomial rays modulo primes with the actual fixed coefficients; the article supplies the accompanying symbolic degree argument.

Every check uses explicit error conditions that remain active under `python3 -O`; the result does not depend on Python assertions remaining enabled. The audits were run both normally and in optimized mode where specified by the combined driver.

The nine-, ten- and fifteen-variable finite auditors use the Python standard library. NumPy is used only for the eleven-variable univariate polynomial-ray checks; the tested version is recorded in the combined receipt. The single command

```
python3 verify_all.py
```

repeats the audits from an extracted archive. To compile this article, run `pdflatex -interaction=nonstopmode -halt-on-error report.tex` twice from its root directory. A `Makefile` provides both targets.

These programs do not exhaust unbounded witnesses, prove the compiler semantics by testing, or settle consistency. Their role is to make the literal data and finite algebra independently reviewable. The new mathematical proofs use explicitly identified inherited compiler and number-theoretic results, all formalizable in the arithmetic available in ZFC. No Lean or Rocq build was performed for the new polynomials.

7.3 Next mathematical improvements worth testing

The support theorem suggests several concrete directions. First, one can search for shorter label vectors, with an independent checker for active-pair isolation unchanged. The present $W = 200$ certificate has no accompanying optimality proof. Its dependence is explicit:

$$\deg F_9 = 162444W + 79768, \qquad \deg F_{10} = 297088W + 147144$$

for this uniform-target construction and the same leading-source conditions. Hence each unit removed from the maximum label saves a known number of degrees. It does not necessarily save arithmetic gates, because addition chains depend on the actual exponent set.

Second, the uniform target gap $2W + 1$ is sufficient but may be wasteful. One could use the actual difference sets of the row supports to place targets more tightly, provided every cross-row interference term is excluded and the off-target carry bound is reproved. Third, an equivalent quadratic source with a different support graph may admit smaller labels even if its number of equations changes. These are optimization questions about a finite certified interface, not a search over unverified compiler semantics.

For the eleven-variable route, the useful new quantity is the relative error between $I\sqrt{G}$ and $c\sqrt{D}$. Retaining such a difference before introducing coarse absolute bounds may permit further combined eliminations. For coefficient size, any smaller coherent tuple must be substituted as a whole and all source magnitude estimates recomputed; the current archive deliberately uses one authenticated tuple throughout. Finally, an explicit formal bridge from the materialized native compiler to the syntactic proof predicate would strengthen certification without changing these arithmetic optimization questions.

A A literal finite recipe for the fixed coefficient tuple

The following restates the package-31 coefficient recipe in a form that can be reconstructed independently of its original builder [3]. All the numbers in this appendix are fixed constants, not additional Diophantine unknowns. The attached literal table contains 943 ordered Wang tiles. Its first two entries are $(2, 435, 3, 1)$ and $(5, 0, 0, 0)$; the remaining entries are sorted lexicographically. Each entry lists west, north, east and south labels, all in $\{0, \dots, 511\}$. The full finite table is part of the exact specification, at `coefficients/upstream/wang943_tiles.json`.

Set

$$\rho = 2^{121}, \quad L_c = 8801, \quad d_c = 121 \cdot 8801 = 1064921, \quad B_c = 2^{d_c}, \quad W_j = 2^{10+2j} \quad (0 \le j \le 43).$$

Write $\mathrm{bit}_j(n)$ for the jth binary digit of a nonnegative integer. If the tile with selector e has labels $x_{e,0}, \dots, x_{e,3}$, put

$$c_e = 1 + \sum_{t=0}^{3} \sum_{j=0}^{9} \mathrm{bit}_j(x_{e,t} + 512) W_{10t+j} + \sum_{j=40}^{43} W_j \qquad (0 \le e \le 942).$$

The additional copy positions are defined using $(o_0, o_1, o_2, o_3) = (1023, 1013, 1003, 983)$:

$$c_{o_t+j} = W_{10t+j} \quad (0 \le t \le 3,\ 0 \le j \le 9).$$

Put

$$(c_{1095}, c_{1096}, c_{2195}, c_{3296}) = (W_{40}, W_{41}, W_{42}, W_{43}), \qquad c_{1063} = c_{1064} = 0.$$

The complete 989-element selector set is

$$\mathcal{E} = \{0, \dots, 942\} \cup \{983, \dots, 992\} \cup \{1003, \dots, 1032\} \cup \{1063, 1064, 1095, 1096, 2195, 3296\}.$$

Define

$$\begin{aligned}
\mu &= 1022 + \sum_{j=0}^{43} W_j, \\
Z_{\mathrm{pad}} &= (2^{119}-1) \sum_{e=8792}^{8798} \rho^e + (2^{27}-1)\rho^{8799}, \\
D_C &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2200} + \sum_{e\in\mathcal{E}} c_e(\rho^{4393-e} + \rho^{5495-e}), \\
D_R &= \rho^{42}, \\
M_C &= B_c - 1 - \sum_{\substack{e\in\mathcal{E}\\ e\neq 1}} \rho^e - 2\rho^{1063}, \\
M_F^{\mathrm{native}} &= 4 + \mu(\rho^{4393} + \rho^{5495}) + \sum_{j=0}^{9} (\rho^{1013+j} + \rho^{1065+j}) + \rho^{2195} + \rho^{3296} + Z_{\mathrm{pad}}.
\end{aligned}$$

The four ports used throughout the article are exactly

$$\boxed{\beta = B_c - 1, \qquad K_0 = D_C + B_c D_R, \qquad M_C = M_C, \qquad M_F = M_F^{\mathrm{native}} + B_c - 1.} \tag{20}$$

In particular, the final port is the *shifted* mask. Substituting a native mask in its place, or mixing coefficients from different packages, would define a different polynomial and invalidate the inherited binding.

A.1 Byte and arithmetic checks

The independent reconstruction uses only this recipe and the literal tile table; it does not import the original builder. It checks the exact integer values against all four supplied little-endian unsigned payloads, both compressed and decompressed SHA-256 hashes, and the dimensions recorded in the binding manifest. It also verifies

$$0 < M_C < \beta, \quad 0 < M_F^{\mathrm{native}} < \beta, \quad M_C \equiv 2 \pmod 4, \quad M_F^{\mathrm{native}} \equiv 4 \pmod 8,$$

and population counts

$$\mathrm{pc}(M_C) = 1064921 - 989, \qquad \mathrm{pc}(M_F^{\mathrm{native}}) = 989.$$

The literal carry threshold is

$$789998487493936316431751375694647416,$$

which lies strictly between 2^{119} and 2^{120}. Finite controller unit residues and the multiplicative orders $10, 12, 52, 60$ modulo $121, 13, 677, 2341$, respectively, are checked as well. These are finite arithmetic checks of the coefficient recipe. The interpretation of arbitrary solution words is the separate inherited compiler theorem.

The archive also materializes all nonliteral fixed leaves introduced by the two packed evaluators: 17 for the nine-positive graph and 16 for the ten-integer graph. Their signed encodings, recipes and hashes are recorded in `fixed/FIXED_LEAVES.json`. Fixed leaves are not counted as variables or charged as variable-dependent arithmetic. The size of a fully expanded polynomial coefficient list is neither materialized nor claimed as an improved metric.

References

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